Table 27. Bangla text challenges and normalisation rules

Challenge	Example	Rule or method
Alternative encodings of nukta letters	য় as one code point or as য + nukta	Unicode NFC (decomposes U+09DC, U+09DD, U+09DF consistently)
Legacy khanda ta	ত + virama + ZWJ instead of ৎ	Map the sequence to U+09CE before removing zero-width characters
Zero-width characters	ZWJ, ZWNJ, soft hyphen inside words	Remove for matching; keep the original for display
Bangla and ASCII digits	২০২৬ and 2026	Fold Bangla digits to ASCII for matching
Inflection	নীতি, নীতির, নীতিসমূহের	Light suffix stripping; index both surface form and stem
Spelling variants and reforms	সরকারী and সরকারি; জাতীয় written with different encodings	Variant list maintained by language editors; NFC first
Code-mixing and romanised Bangla	budget বরাদ্দ; romanised queries such as bajet	Bilingual glossary; transliteration candidates in later versions
OCR errors in conjuncts	ক্ষ, ঞ, ন্ত্র misread	OCR quality score per page; human correction queue
Bangla calendar dates	Dates given in the Bangla calendar	Store both Gregorian and Bangla dates where given
Name transliteration	Ministry and place names in several spellings	Authority list of names with variants

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